Peer Review File

Manuscript Title: Catalogue of Flat Band Stoichiometric Materials

Reviewer Comments & Author Rebuttals

Reviewer Reports on the Initial Version:

Referee #1 (Remarks to the Author):

Report on the manuscript « Catalogue of Flat Band Stoichiometric Materials » by N. Regnault et al.

Key results, Validity and Originality

This article builds on the recent development of topological quantum chemistry by A. Bernevig, N. Regnault and collaborators. Its specific goal is to identify promising candidate materials displaying so-called flat energy bands, and potentially displaying exotic strongly interacting phases. The study of such states of matter originates from the discovery of fractional quantum Hall states in the 80 's, and was seriously revitalized by the recent study of twisted bilayer graphene. Identifying materials in which physics analogous to twisted bilayer graphene can take place is a natural and major objective of the material and quantum chemistry community. The present manuscript is the result of the successful program undertaken by the authors. It is heavily based on a method, coined a S-matrix approach, developed in a companion manuscript, and the material database developed by a group around some of the authors. The key results are thus a list of candidates materials which a promising platform to look for exotic interacting states of matter. I have no doubt that such a catalog will be of great interest in the following years for exploratory studies of such phases. We can only hope for new discoveries based on this catalog.

In the recent years, the work by A. Bernevig and co-workers has scrambled the domain of numerical quantum chemistry, bringing topology as an essential discriminating filter to identify exotic solid state phases. The results of this manuscript are an attempt to extend the scope of this program towards strongly interacting states of matter. The methods are original, the results stimulating. The methods are well presented and convincing, although I cannot comment on the validity of the numerical approches which are beyond my domain of expertise.

Data & methodology

I have some concerns about the presentation of results, and in particular the huge lists of data provided in the form of table III, V, VII, VIII.

To my eyes, they look so old fashion and 20th century style. An online datable, such as the one provided by the authors, is immensely more practical. Hence I question the real use of these long written tables (besides setting some records). Maybe the sustainability of the data is at sake here. I'll leave it to the editor to decide on this point.

Suggested improvements

1. In this article on a subject of strongly interacting phases targeting a broad audience, no mention to high Tc superconductors is made. I believe that it is the keystone in the community of strong coupling phases. A contrast (or similarity ?) with the expected phases in flat band materials would be welcome.

2. A great part of the method used by the authors consists in identifying crystals which possess, even if approximately, a bipartite (sub)lattice with hopping between its two components. This is just a geometrical implementation of the chiral symmetry, one of the three symmetries underlying the ten-fold classification of gapped topological phases (see e.g. Topological Insulators and Topological Superconductor by A. Bernevig). It was initially discussed in

Bill Sutherland, "Localization of electronic wave func- tions due to local topology," Phys. Rev. B 34, 5208 (1986)

E. H. Lieb, « Two theorems on the Hubbard model », Phys. Rev. Lett. 62, 1201 (1989).

and more recently, in the context of topological gapped phases, in e.g.

Shinsei Ryu et al., Topological insulators and superconductors: tenfold way and dimensional hierarchy, 2010 New J. Phys. 12 065010.

Many implementations in quantum simulators (cold-atoms, light waveguides, polariton lattices) or in mechanical systems possess this symmetry.

I think that the relation to the chiral symmetry should be mentioned.

Moreover, does the reliance on this (approximate) geometrical symmetry to seek flat bands has consequences on the topological properties of the single particle eigenstates of the flat band ?

Finally, chiral symmetry is usually only a low energy approximation. Indeed the authors discard hopping terms smaller than an ad-hoc cutoff. Does this approximation have any consequences on the strength of interactions to consider and the associated nature of interacting states ?

3. On p.5, five candidate materials are selected and said to be paramagnetic instead of Mott insulators. It is unclear why this is an advantage. The authors should clarify this point.

4. Flat bands are a single particle notion. Yet the authors have performed Density Functional Theory (DFT) analysis of the huge list of materials. DFT is known to incorporate effects of (strong) electronic interactions. What are the implications of this ?

Does that mean that interactions are ineffective in the candidate materials identified by the authors ? Or that DFT did not capture the potential interacting phases and associated gaps ?

This manuscript would constitute a real milestone is a complementary method, such as DMFT and its various refinements, was use in the five material candidates to precise the nature of possible interacting instabilities. Given the expertise of the authors, this should not be a major obstacle.

References

Citations [3-6] do not refer to the historical papers on the Fractional Quantum Hall State ! Although I agree that recent work on topological interacting phases is stimulating, we cannot ignore the history of the quantum Hall effect which predates the study of fractional Chern insulators. The authors should do justice to the historical papers of this field, such as the pioneering papers of Robert Laughlin, Bertrand Halperin and the experimental papers of Horst Störmer, and Daniel Tsui.

Referee #2 (Remarks to the Author):

In this work, Regnault et al report the results from their systematic search of stoichiometric materials with relatively flat bands over some segments of the Brillouin zone. The main output of the work is provided as an online database and also in the appendices. The present manuscript is devoted to introducing the search algorithms and providing more detailed elaboration on five representative materials they found, together two more in the Appendix. My evaluation on the manuscript is divided into two aspects. First, whether or not the list of proposed materials will be of interest to the community. Second, if the approach taken by the authors is novel and could lead to new developments.

For the first part, there is no doubt that there is immense scientific interest in materials with flat electronic bands not coming from trivially decoupled atomic orbitals. The present generation of materials studied is mostly based on a layered kagome motif. The database introduced by the current work can indeed expand the possible directions in which one looks for new flat-band materials. The reported materials are not perfect, but they could be interesting especially when chemically doped to bring the flat bands to the Fermi energy. That said, one has to bear in mind that the list here is curated out of an earlier work by some of the authors. In a sense, the authors have now added an additional “flat band” filter to their existing data. This will be useful for those who would want to look for flat band materials of any sort in any system, excluding the trivially atomic ones. However, if one starts with a specific material family in mind, one could query the previously published TQC database and discover for themselves that the electronic bands contain flat segments. In this regard, the present work adds value by excluding the trivially atomic flat bands.

For the second part, the authors used three main strategies for identifying interesting flat bands. The first is a relatively crude post-processing of the data in the TQC database to identify flat segments of electronic bands and peaks in density of states. The second is a geometric approach, which automatically identifies near-realization of three well known flat band geometries: kagome, pyrochlore, and Lieb lattices. Such automatic recognition is not a trivial task, given the large number of space group and Wyckoff combinations that could correspond to the same geometry, and also the practical need to identify approximate realization of these lattices. This space-group method can be viewed as crystallography at its finest.

The last approach is centered around what the authors call the “S-matrix method”, which attempts to generalize the flat bands in split lattices like the Lieb lattice to a more general setup. The theoretical details of this last method are discussed in a separate paper (arXiv:2106.05272), and the present work contains a brief introduction in Appendix E. The gist of this method is the observation that if a system is bipartite and only inter-sublattice couplings are considered, then the Hamiltonian is block off-diagonal. Furthermore, if the two subsystems have different sizes then the the block matrix of electron coupling, called S, will admit either left or right zero modes. The last part is an elementary fact in linear algebra, and as the authors pointed out it does not rely on any further simplification like use of s orbitals or absence of spin orbit coupling. Although the authors discussed some slight relaxation, like the possibility of more general intra-sublattice coupling in the smaller sublattice and also a constant energy shift in the large sublattice, such approximate bipartition is still a very stringent requirement, as intra-sublattice coupling can easily lead to substantial deviation to the limit for which the method is applicable.

Therefore, the real question for applying this “S-matrix” approach to material research is the difficulty in realizing the restriction. In practice, the authors impose a distance cutoff to the electron coupling range, and then look for materials for which all the remaining couplings are inter-sublattice. Conceptually, this is a reasonable criterion to impose, but it is unclear to me if an elaborated theory is required to support this simple criterion. Practically, one can also ask if the incorporation of this “S-matrix” method adds substantially to the authors’ results, given the Lieb lattice is a special case of the split lattices, which are special cases of the bipartite lattices. From the last column of Table I, if we add up the entries flagged as Kagome (54.37%), Pyrochlore (8.11%), Lieb (15.91%), and None

(26.13%), we arrive at 104.52%. Presumably there is no overlap between the None category and the other three, and it is unclear how much overlapping there is between the three named lattice types. Without this information, it is unclear how important the “S-matrix" based bipartite and split lattice methods are for obtaining the authors’ results (after excluding the Lieb lattice).

Author Rebuttals to Initial Comments:

Reply to the first Referee

Key results, Validity and Originality

This article builds on the recent development of topological quantum chemistry by A. Bernevig, N. Regnault and collaborators. Its specific goal is to identify promising candidate materials displaying so-called flat energy bands, and potentially displaying exotic strongly interacting phases. The study of such states of matter originates from the discovery of fractional quantum Hall states in the 80's, and was seriously revitalized by the recent study of twisted bilayer graphene. Identifying materials in which physics analogous to twisted bilayer graphene can take place is a natural and major objective of the material and quantum chemistry community. The present manuscript is the result of the successful program undertaken by the authors. It is heavily based on a method, coined a S-matrix approach, developed in a companion manuscript, and the material database developed by a group around some of the authors. The key results are thus a list of candidates materials which a promising platform to look for exotic interacting states of matter. I have no doubt that such a catalog will be of great interest in the following years for exploratory studies of such phases. We can only hope for new discoveries based on this catalog.

In the recent years, the work by A. Bernevig and co-workers has scrambled the domain of numerical quantum chemistry, bringing topology as an essential discriminating filter to identify exotic solid state phases. The results of this manuscript are an attempt to extend the scope of this program towards strongly interacting states of matter. The methods are original, the results stimulating. The methods are well presented and convincing, although I cannot comment on the validity of the numerical approches which are beyond my domain of expertise.

We thank the referee for her/his positive evaluation of our work and saying "such a catalogue will be of great interest in the following years for exploratory studies of such phases".

The validity of our method has been confirmed by several successful material predictions. Our current work and catalogue of flat-band materials are based on the topological material database (https://www.topologicalquantumchemistry.com/). This database was built using first-principles calculations and the Topological Quantum Chemistry (TQC) theory [*Nature 547 (7663), 298-305 (2017), Nature 566 (7745), 480-485 (2019), arXiv:2105.09954(2021)*]. The success of TQC is marked by the new topological materials it found. For example, the first higher-order topological insulator Bi was predicted by the TQC method and successfully proved in the experiments [*Nature physics 14 (9), 918-924 (2018)*]; also, the predicted hinge states in Bi_4Br_4 have been measured with ARPES experiments [*2D Mater. 6 031004(2019); Nature Materials 20, 473–479 (2021)*]. All these materials were first available in the TQC high-throughput calculations and material database (www.topologicalquantumchemistry.com/). Using TQC, the higher-order hinge arcs in Cd_3As_2, KMgBi, and -PtO_2 [*Nature Communications 11, 627 (2020)*] were also predicted and subsequently measured in experiments [*Phys. Rev. Lett. 124, 156601 (2020)*], and researchers have predicted the fragile phase in twisted bilayer graphene [*Phys. Rev. Lett. 123,036401 (2019)*], the TCI phase in Pt_2HgSe_3 [*Phys. Rev. Materials.3.074202 (2019)*], the axionic CDW phase in the Weyl Semimetal $(TaSe_4)_2I$ [*Nature 575, 315–319(2019), Nat. Phys. 17, 381-387 (2021)*], and the dark matter detectors $Ag_2Au(Se_2,Te_2)$ [*J. Phys. Mater. 3 014001(2019)*]. Hence, we strongly believe that the underlying numerical results and methods for the topological material database are e□cient, a statement that thus transposes to our catalogue of flat-band materials.

Data & methodology

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To my eyes, they look so old fashion and 20th century style. An online datable, such as the one provided by the authors, is immensely more practical. Hence I question the real use of these long written tables (besides setting some records). Maybe the sustainability of the data is at sake here. I'll leave it to the editor to decide on this point.

We agree that these long tables might look outdated. This is why we also provide full fledged websites with search engines. Nevertheless through collaborations and feedback, we have realized that some people are more comfortable with tables than websites. Some of them actually print these tables. Moreover, it provides a convenient way for data preservation that is mandatory for publication in scientific journals.

The tables shall be seen as a record and an addition to the websites. As such, they are not a substitute to any online feature. Their location in the supplementary material is chosen to avoid disrupting the text flow. All of them have direct links to our various websites, so readers can have the best of both worlds.

Suggested improvements

1. In this article on a subject of strongly interacting phases targeting a broad audience, no mention to high Tc superconductors is made. I believe that it is the keystone in the community of strong coupling phases. A contrast (or similarity ?) with the expected phases in flat band materials would be welcome.

We thank the referee for pointing out high Tc superconductors as relevant for a broad audience. Our database, as well as the lists of curated and best representative materials, informs the reader if a material is known to be a superconductor (including high Tc), based on the NIMS Materials Database (https://mits.nims.go.jp/en/). Additionally, this point is now explicitly written at the end of the introduction *"Our classification and predictions take into account the different flat band natures, including their topological character, and our database is coupled to the Materials Project and* the NIMS Materials Database, *providing information about the magnetic and superconducting properties (including high Tc) of the candidate materials*." Moreover, we have shortened the introduction and put more emphasis on the exotic phases that might emerge from flat topological bands.

2. A great part of the method used by the authors consists in identifying crystals which possess, even if approximately, a bipartite (sub)lattice with hopping between its two components. This is just a geometrical implementation of the chiral symmetry, one of the three symmetries underlying the ten-fold classification of gapped topological phases (see e.g. Topological Insulators and Topological Superconductor by A. Bernevig). It was initially discussed in

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and more recently, in the context of topological gapped phases, in e.g.

Shinsei Ryu et al., Topological insulators and superconductors: tenfold way and dimensional hierarchy, 2010 New J. Phys. 12 065010.

Many implementations in quantum simulators (cold-atoms, light waveguides, polariton lattices) or in mechanical systems possess this symmetry.

I think that the relation to the chiral symmetry should be mentioned.

We thank the referee for pointing this out. We have added several sentences discussing the relationship between flat bands and the chiral symmetry in the Methods section of the revised manuscript. The related references are also cited.

We emphasize that the S-matrix method used in this presented work, as well as the BR subduction rule used in our accompanying work [*arXiv:2106.05272 (2021)*], goes beyond the (approximate) chiral symmetry. In general, a bipartite lattice consists of two sublattices L and L', which have N_L and $N_{L'}$ ($<N_L$) orbitals per unit cell, respectively. Its Hamiltonian matrix is,

$$H_k = \begin{pmatrix} A_k & S_k \\ S_k^\dagger & B_k \end{pmatrix}$$

where A_k (B_k) is a $N_L * N_L$ ($N_{L'} * N_{L'}$) Hermitian matrix denoting the intra-sublattice hopping inside the L (L') sublattice and S_k denotes the hopping matrix between sublattices L and (L'). The presence of the chiral symmetry in a bipartite lattice forbids any intra-sublattice hopping terms and hence $A_k = B_k = 0$. Then, as has been proved in Appendix E1 and also in our accompanying work [*arXiv:2106.05272 (2021)*], H_k contains at least $N_L - N_{L'}$ zero modes in the whole k-space giving rise to $N_L - N_{L'}$ perfectly flat bands, as well as $2N_{L'}$ dispersive bands coming in pairs related by the chiral symmetry.

A similar argument for the existence of flat bands holds for a generalized bipartite lattice, which includes intra-sublattice coupling and breaking of the chiral symmetry. To see this, we assume that A_k has a momentum independent eigenvalue ($E=a$) with degeneracy n_a. If $N_{L'} < n_a \leq N_L$, then H_k has at least $n_a - N_{L'}$ perfectly flat bands of energy $E=a$ irrespective of B_k. (This statement is proved in the Appendix I4 of the accompanying work arXiv:2106.05272). Compared to the chiral case, the presence of intra-sublattice hopping in the smaller sublattice (L') gaps the accidental band touching points that exist in a chiral case. On the other hand, the eigenstates of these $n_a - N_{L'}$ flat bands as well as the stable touching points remain unperturbed in the generalized bipartite lattice. (More details can be found in Appendix I4 of Ref. [arXiv:2106.05272]).

Moreover, does the reliance on this (approximate) geometrical symmetry to seek flat bands has consequences on the topological properties of the single particle eigenstates of the flat band ?

As pointed out in the Sec II B ***"Geometric frustration in (line-graph and bipartite) lattices is known to give rise to exact FTBs"***, In fact, many interference-originated bands are topologically nontrivial. Some examples were discussed in detail in some recent works, e.g. *Phys. Rev. B 78, 125104 (2008), Phys. Rev. B 99, 125122 (2019), Phys. Rev. Lett. 125 (26), 266403 (2020), Phys. Rev. Research.2.043414 (2020)*. A systematic theoretical framework based on topological quantum chemistry to classify and diagnose the flat-band topology is discussed in an accompanying work [*arXiv:2106.05272 (2021)*]. In the *S*-matrix method, in order for the band structure to possess flat bands, the two sublattices should in general have different numbers of degrees of freedom. Let us say the two sublattices *L* and *L', which have* N_L *and* $N_{L'}$ $(<N_L)$ orbitals per unit cell, respectively. Then there should be N_L-$N_{L'}$ flat bands at zero energy. Moreover, if the larger and smaller sublattices form the band representations *BR* and *BR'*, respectively, the representation formed by the flat bands is given by ***BR-BR'***. (In topological quantum chemistry, the terminology band representation refers to a representation formed by local orbitals. In other words, band representations are topologically trivial insulators.) If ***BR-BR'*** is not consistent with any band representation, the flat bands must be topological. Examples of the BR subduction rule can be found in [*arXiv:2106.05272 (2021)*].

As explained above, the *S*-matrix method, as well as the BR subduction rule, goes beyond the (approximate) chiral symmetry. Starting with a bipartite lattice satisfying the chiral symmetry, the flatness and topology of the flat bands will persist even if intra-sublattice hoppings (which break the chiral symmetry) are added, provided that the total Hamiltonian is still a so-called *"generalized bipartite crystalline lattice"*

Finally, chiral symmetry is usually only a low energy approximation. Indeed the authors discard hopping terms smaller than an ad-hoc cutoff. Does this approximation have any consequences on the strength of interactions to consider and the associated nature of interacting states ?

We thank the referee for this comment. In fact, there is no *ad-hoc cutoff* adopted in the DFT calculations. As shown case by case of the six types of materials in Appendices F1-F6, the first-principles calculations without any *ad-hoc cutoff* show almost perfect flat bands around the chemical potential. In the no-free parameter Wannier tight-binding Hamiltonian of these materials, we find the kinetic hopping parameters beyond a certain distance are negligible, and the hoppings within the cutoff distance satisfy our *S*-matrix conditions. Hence even when the negligible hoppings are considered in the model, the flat bands will only become slightly dispersive or even persist to be perfectly flat. If present, the negligible dispersion won't change the interacting physics since the interaction has a much larger energy scale.

3. On p.5, five candidate materials are selected and said to be paramagnetic instead of Mott insulators. It is unclear why this is an advantage. The authors should clarify this point.

We thank the referee for this comment. In the main text, we only showcase the materials that have flat bands and non-ferromagnetic ground state because all the first-principles calculations on the material database are paramagnetic (i.e., without the on-site spin polarization) and in a single-particle (mean field) approximation. We tagged materials reported as magnetic on the Material Project Database or experimentally in the tables of Appendix H, but we did not perform the high-throughput magnetic calculations and topological classifications for their magnetic ground states. In the revised manuscript, we clarified this point in the first paragraph of Sec III B of the main text, i.e., "*All of the five representative flat-band materials are chemically realistic, experimentally paramagnetic and not Mott insulators, which is consistent with our paramagnetic calculations*".

4. Flat bands are a single particle notion. Yet the authors have performed Density Functional Theory (DFT) analysis of the huge list of materials. DFT is known to incorporate effects of (strong) electronic interactions. What are the implications of this ?

Does that mean that interactions are ineffective in the candidate materials identified by the authors ? Or that DFT did not capture the potential interacting phases and associated gaps ?

We thank the referee for this valuable comment. In the previous works, *i.e., [Nature 547, 298-305 (2017), Nature 566, 480-485 (2019)]* and this present manuscript, we have applied the GGA exchange-correlation potential and enforced the paramagnetic phase in the DFT calculations. The GGA potential usually cannot capture orbital-selected interactions, like Hubbard U for *3d, 4d,* and *4f* electrons. Therefore, for paramagnetic materials without correlated atoms (with *3d, 4d,* or *4f* electrons), one can think that the interaction effect is already incorporated at a mean-field level and the flat band is a feature of the mean-field Hamiltonian. However, for magnetic materials or materials with correlated electrons, one can regard the band structures in our work as spectra of the "single-particle" Hamiltonians. Spin splitting and/or Hubbard U or Hund's interactions should be included to describe magnetism and/or strong correlation physics.

In the newly added Appendix (Appendix G of the revised Supplementary Materials), we picked four flat-band materials which are ferromagnetic in experiments. We performed DFT calculations for the magnetic and non-magnetic phases using the GGA+U method (which included the on-site Hubbard U value of '*d*' or '*f*' electrons in the mean-field approximation). By comparing the total energy of the non-magnetic and magnetic phases for different U values (varying from 0 to 5 eV) with and without SOC, we found that the magnetic phases always have lower total energies. The theoretically obtained magnetic moments of these four materials are in agreement with the experimental results. Hence, their magnetic ground states can be captured by the GGA+U calculation. We also find that the flat bands obtained in the paramagnetic calculations, displayed in the results section of the manuscript, remain in the ferromagnetic calculations.

This manuscript would constitute a real milestone if a complementary method, such as DMFT and its various refinements, was used in the five material candidates to precise the nature of possible interacting instabilities. Given the expertise of the authors, this should not be a major obstacle.

We thank the referee for the suggestion to use alternative/complementary methods to test the many-body interacting ground states versus the experimental ones. We used GGA +U to uncover the magnetic ground state of four compounds in the newly added Appendix (Appendix G of the revised Supplementary Materials). The theoretical results match the experimentally published structures well. Because the structures are ferromagnetic and the GGA+U calculation clearly converges fast to the correct experimental ground state, we expect further alternative methods to give the same result, at least for the experimentally known strong ferromagnetic compounds. Specifically, in this newly added Appendix, we picked four flat-band materials i.e. SrRuO3[ICSD 69360], Rb2MnCl6[ICSD 9347], Ba2MnReO6[ICSD 4169], and NiMnSb[ICSD 54255] which are ferromagnetic in experiments and did the DFT calculations. The results show that the GGA+U method can predict the ground state of the magnetic ground state. The magnetic band structures are shown in figures 23-26 in the revised Supplementary Materials. We find that, in the paramagnetic calculations, when the Hubbard U values are added the flat bands remain at (or close to) the Fermi level. When we turn to the ferromagnetic calculations, for different U values, the flat bands are no longer located at the Fermi level but split into two sets of flat bands near the Fermi level. Thus, the flat band features obtained in the paramagnetic calculations, displayed in the results section of the manuscript, remain in the ferromagnetic calculations, but their energies change. This type of results justifies why we are confident that a high-throughput search for flat bands based on paramagnetic ab-initio calculations (without SOC) is both meaningful and relevant even for potentially (anti-)ferromagnetic materials.

References

Citations [3-6] do not refer to the historical papers on the Fractional Quantum Hall State ! Although I agree that recent work on topological interacting phases is stimulating, we cannot ignore the history of the quantum Hall effect which predates the study of fractional Chern insulators. The authors should do justice to the historical papers of this field, such *as the pioneering papers of Robert Laughlin, Bertrand Halperin and the experimental papers of Horst Störmer, and Daniel Tsui.*

We apologize for this mistake. We agree that we should have cited there the pioneer work in the field of fractional Quantum Hall effect. We have fixed the citations in the manuscript.

Reply to the second Referee

In this work, Regnault et al report the results from their systematic search of stoichiometric materials with relatively flat bands over some segments of the Brillouin zone. The main output of the work is provided as an online database and also in the appendices. The present manuscript is devoted to introducing the search algorithms and providing more detailed elaboration on five representative materials they found, together two more in the Appendix. My evaluation on the manuscript is divided into two aspects. First, whether or not the list of proposed materials will be of interest to the community. Second, if the approach taken by the authors is novel and could lead to new developments.

For the first part, there is no doubt that there is immense scientific interest in materials with flat electronic bands not coming from trivially decoupled atomic orbitals. The present generation of materials studied is mostly based on a layered kagome motif. The database introduced by the current work can indeed expand the possible directions in which one looks for new flat-band materials.

We thank the referee for her/his evaluation of our work stating that"*The database introduced by the current work can indeed expand the possible directions in which one looks for new flat-band materials*". Indeed, the method given in our manuscript is exhaustive and general. The flat bands that attracted abundant scientific interest in the physics and materials communities do not originate from decoupled atomic orbitals. At present, almost all the known materials that possess flat bands and the models predicting the flat bands can be explained using our S-matrix theory. (See Section II of Supplementary Materials of *arXiv:2106.05272 (2021)*.)

The reported materials are not perfect, but they could be interesting especially when chemically doped to bring the flat bands to the Fermi energy.

We agree with the referee that there are several materials with flat bands away from the Fermi level and hence chemical doping is needed. Nevertheless, as tabulated in Appendix G3, we select 345 best material candidates (corresponding to 949 ICSD entries) that have flat bands on or very close to the Fermi level. These materials do not need (or just need a slight) chemical doping to have the Fermi level run through the flat bands.

That said, one has to bear in mind that the list here is curated out of an earlier work by some of the authors. In a sense, the authors have now added an additional "flat band" filter to their existing data.

As we point out in the manuscript, we indeed rely on the latest version of the TQC database *[arXiv:2105.09954(2021)]*. With the exception of the projected density of states (DOS), the band structure, lattice information and the topological properties were already available on the TQC website. But the work presented in this manuscript goes beyond simple addition of a "flat band" filter. To perform a high-throughput search of relevant flat-band materials, we have developed several tools to detect features in the band structures, DOS and lattice structures, as detailed at length in the Supplementary Materials. The whole data had to be processed through these tools, and organized to discover the most interesting and promising compounds. The website might look like a fully-tunable filter on the TQC database to the end user, but this impresion should not overshadow the work and complexity lying under the hood.

This will be useful for those who would want to look for flat band materials of any sort in any system, excluding the trivially atomic ones. However, if one starts with a specific material family in mind, one could query the previously published TQC database and discover for themselves that the electronic bands contain flat segments.

Technically, the referee is right: someone could look for a *given* compound, check if it has non-trivial flat bands and compute if these flat bands emerge from a remarkable sublattice. Still, it would not be possible to search through *all the materials* without the tools, data processing, and concepts that we have developed in this manuscript and the accompanying paper on S-matrix method. Of course, a person motivated enough could theoretically perform a manual check of the individual 35k entries (a several months of tedious work) to look for new candidates. With the new website that we provide to the community, it only takes a few seconds. We are sure that the referee appreciates the value of such tools.

In this regard, the present work adds value by excluding the trivially atomic flat bands.

We would like to convince the referee this is far from being the only added value. Besides the tools described in the reply to the last question, in this work we also explained the physical origins of the flatness and why they tend not to be atomic flat bands. We point out that the method also shows why flat bands by interference tend to be topological.

(i) A systematic theoretical framework for flat bands, i.e., the *S*-matrix method, is developed in the accompanying work *[arXiv:2106.05272 (2021)]* and briefly discussed in the Appendix E of the present work. In general, if there are two sublattices connecting to each other and the hoppings within each sublattice either vanish or satisfy the so-called "*generalized bipartite lattice*" condition, there will be exact flat bands in the middle of the spectrum. The number of flat bands depends on the number of orbitals in the two sublattices and the rank of the hopping matrix.

(ii) In the present work, we have applied the *S*-matrix theory to explain the flatness found in the band structures in the database. In Appendices F1-F6, we demonstrate the *S*-matrix theory in six flat-band materials by explicitly constructing the effective tight-binding models. In Appendix H, we exhaustively tabulated all materials with bipartite lattice structures or line-graph lattice structures. (Flat bands in line-graph can also be explained by the *S*-matrix.) We categorized the materials into Kagome, Lieb, pyrochlore, and generic bipartite lattices. A large number of materials (3756 ICSDs) with such special lattices are found to possess flat bands.

(iii) As explained in the reply to the first referee, the topology of the flat bands can be easily determined by a band representation subduction rule. (In topological quantum chemistry, the term *band representation* refers to a representation formed by local orbitals. In other words, band representations are trivial insulators.) We suppose that the larger sublattice in the S-matrix theory forms a band representation ***BR*** and the smaller sublattice forms a band representation ***BR'***, then the representation of the flat bands will be ***BR-BR'***. There is a high chance that ***BR-BR'*** is not consistent with any atomic orbitals in the given system. ***BR-BR'*** may even be a topological band if it cannot be written as a sum of elementary band representations.

For the second part, the authors used three main strategies for identifying interesting flat bands. The first is a relatively crude post-processing of the data in the TQC database to identify flat segments of electronic bands and peaks in density of states. The second is a geometric approach, which automatically identifies near-realization of three well known flat band geometries: kagome, pyrochlore, and Lieb lattices. Such automatic recognition is not a trivial task, given the large number of space group and Wyckoff combinations that could correspond to the same geometry, and also the practical need to identify approximate realization of these lattices. This space-group method can be viewed as crystallography at its finest.

We are pleased to see the referee appreciates the complexity of identifying these remarkable lattices. We hope our effort that she/he emphasizes, will be helpful to a large community, beyond the one of quantum materials.

The last approach is centered around what the authors call the "S-matrix method", which attempts to generalize the flat bands in split lattices like the Lieb lattice to a more general setup. The theoretical details of this last method are discussed in a separate paper (arXiv:2106.05272), and the present work contains a brief introduction in Appendix E. The gist of this method is the observation that if a system is bipartite and only inter-sublattice couplings are considered, then the Hamiltonian is block off-diagonal. Furthermore, if the two subsystems have different sizes then the the block matrix of electron coupling, called S, will admit either left or right zero modes. The last part is an elementary fact in linear algebra, and as the authors pointed out it does not rely on any further simplification like use of s orbitals or absence of spin orbit coupling. Although the authors discussed some slight relaxation, like the possibility of more general intra-sublattice coupling in the smaller sublattice and also a constant energy shift in the large sublattice, such approximate bipartition is still a very stringent requirement, as intra-sublattice coupling can easily lead to substantial deviation to the limit for which the method is applicable.

We would like to convince the referee that the argument "*such approximate bipartition is still a very stringent requirement, as intra-sublattice coupling can easily lead to substantial deviation to the limit for which the method is applicable*" does not hold here. First, we point out that almost all the known mechanisms of flat bands can be thought of as special types of the *S*-matrix theory. (See Section II of Supplementary Materials of arXiv:2106.05272 (2021).) In this sense, the *S*-matrix is already the most generic and powerful theory framework for flat bands. As far as we know, there is no theory that can guarantee flat bands and does not require the chiral symmetry or any "*stringent*" conditions on the hoppings.

Second, whether the S-matrix is a "stringent" condition should be answered by the actual material search. In fact, many materials are indeed well described by the *S*-matrix theory. In Appendices F1-F6, we show how the *S*-matrix theory can successfully explain flat bands in six example prototypical materials by explicitly constructing/extracting the effective tight-binding models. In Appendix H, we find that materials with bipartite lattice structures or line-graph lattice structures (special cases of the *S*-matrix theory) have higher chances to possess flat bands.

Therefore, the real question for applying this "S-matrix" approach to material research is the di□culty in realizing the restriction.

As shown for the six materials given in Appendix F, the results obtained from the S-matrix method match well with those from the DFT calculations. We picked these six materials as typical examples of compounds that can have flat bands. The materials listed at the end of each subsection of Appendix F 1~6 are analogous with these six materials, and their corresponding flat bands can be explained by the S-matrix method (sometimes the same S matrices). Moreover, the S-matrix method can be applied to all flat bands that are not formed by the decoupled atomic orbital. Thus, realizing the S-matrix requirements in real materials is feasible.

In practice, the authors impose a distance cutoff to the electron coupling range, and then look for materials for which all the remaining couplings are inter-sublattice. Conceptually, this is a reasonable criterion to impose, but it is unclear to me if an elaborated theory is required to support this simple criterion.

We thank the referee for this comment. In principle, the distance between two orbitals affects the strength of coupling between them (if the coupling is not symmetry-guaranteed to be zero). As we show in the Appendix F1 of the revised Supplementary Materials on WO3 as an example, the S-matrix theory captures the main coupling (inter-sublattice coupling 1.654 eV) that forms flat bands. The neglected couplings (i.e., the intra-sublattice couplings -0.222 eV and 0.179 eV) are much weaker than the inter-sublattice coupling. The assumption that intra-sublattice couplings vanish is in good agreement with the DFT results.

Practically, one can also ask if the incorporation of this "S-matrix" method adds substantially to the authors' results, given the Lieb lattice is a special case of the split lattices, which are special cases of the bipartite lattices. From the last column of Table I, if we add up the entries flagged as Kagome (54.37%), Pyrochlore (8.11%), Lieb (15.91%), and None (26.13%), we arrive at 104.52%. Presumably there is no overlap between the None category and the other three, and it is unclear how much overlapping there is between the three named lattice types. Without this information, it is unclear how important the "S-matrix" based bipartite and split lattice methods are for obtaining the authors' results (after excluding the Lieb lattice).

As the referee pointed out, there is no overlap between the category "None" and the other categories. But indeed, we do have overlaps between the remarkable sublattices. A trivial example is the overlap between the "Kagome" and the "Pyrochlore" categories, since every pyrochlore lattice detection will also trigger a kagome lattice detection. Moreover, some compounds might host both a kagome lattice and, e.g., a bipartite lattice. One such example is WO_3 in SG 221 (ICSD 108651) and discussed in Appendix F

https://www.topologicalquantumchemistry.fr/flatbands/index.html?ICSD=108651.

The online version of the best cases (the last column pointed out by the referee) is available here

https://www.topologicalquantumchemistry.fr/flatbands/bestflatband.html.

There, it is easy to observe this is not an isolated case. To be more quantitative, there are 247 ICSD entries among the 949 ICSD entries in the list of best cases that have both a Kagome and a bipartite sublattice. We have added tables in Appendix C providing the number of cases where a material hosts both a Kagome/pyrochlore and a bipartite sublattices.

Still we stress that the Kagome/pyrochlore sublattices can also be viewed as a limit case in the S-matrix formalism. In principle for any given material, we could perform the same S-matrix analysis that we did in Appendix F to associate each flat-band segment to one type of sublattice. While a systematic (high throughput) S-matrix description is out of reach, Appendix F already covers more than 50 materials. The above mentioned WO_3 is a perfect example. The flat bands arise from the bipartite lattice made of the W and O atoms. The Kagome lattice made only from the O atoms does not contribute to the flat segment but is still automatically detected by our algorithm.

Reviewer Reports on the First Revision:

Referee #1 (Remarks to the Author):

The authors have revised mostly the supplementary material to answer questions by both referees. In particular I appreciate the addition of results on ferromagnetic flat band materials obtained using a GGA+U method, as well as the brief discussion of flat band in (effectively) chiral symmetric lattice models in the Method section.

I am convinced that this paper will have substantial impact in search for various exotic phases in flat band materials and I recommend its publication.

Referee #2 (Remarks to the Author):

I thank the authors for the reply to the comments and for the efforts to revise the manuscript. In particular, I appreciate the authors' effort to clarify the relevance of the "S-matrix"-based method in the identification of good flat band materials. As the authors wrote, "whether the S-matrix is a "stringent" condition should be answered by the actual material search." I totally agree with this point, and as I noted in the original report it is important to clarify the relevance of the "S-matrix" formalism in the material search.

For instance, suppose one simply goes through the existing database using the DOS and flat-band segment checks and the crystallographic screening of the three remarkable lattice types (kagome, pyrochlore, and Lieb), how many of the 949 "best" flat band materials will remain in the "None" category, i.e., will not have an identified physical origin for the emergence of flat bands? Currently, we know that 949 (all under "best) -248 (None) = 701 of the materials can be explained. If, for instance, one finds that kagome + pyrochlore + Lieb accounts for 700 of these entries, then one could say that the S-matrix formalism has helped to explain the physical origin of flat bands in 1 extra material through the bipartite lattice search. This is a non-zero yield, but one would likely agree that it is not very significant (1/701). (Of course, this interpretation did not take into account cases like WO3, which would have been flagged under the kagome category although, as explained by the authors, the kagome sub-lattice has nothing to do with the flat bands near the Fermi level. But isolating such cases will require some more in-depth analysis of the identified materials.)

I believe the conclusion of the manuscript will be significantly strengthened if the authors can simply quantify the explanatory power of the "S-matrix theory" by providing a direct answer to the following question: how many entries have been "rescued" from the None category through the bipartite lattice search? In the first version of the manuscript, it was difficult to even estimate the yield of the S-matrix discussion because of the overlapping between the three named lattice types.

In the revised manuscript, the authors have presented data concerning the overlaps in Table III of Appendix C. For instance, it is interesting to realize that of the 151 Lieb lattices in the "best" column, 135 of them are simultaneously identified under the kagome category. This accounts for a good fraction of the overlap between the three named lattice types. If we assume there is no overlap between kagome and pyrochlore, the numbers add up to be 516 (kagome) + 77 (pyrochlore) + 16 ("pure" Lieb which is neither kagome nor pyrochlore) = 609. This suggests the remaining 701-609 = 92 materials are really explained by the bipartite lattice search. If that is a fair estimate, it will be a decent yield (92/701 of the explained material = 13%), and the readers will likely be much more convinced that the S-matrix theory has additional explanatory power beyond providing a framework for the interpretation of the well-known flat-band lattice types.

Lastly, a comment on terminology: as the authors definitely know, the term "S-matrix" has a widespread, well-accepted meaning in physics. The authors' decision to call their framework "S-matrix" seems to come from the rather ad hoc choice of using the symbol S to denote the off-diagonal entry in the Bloch Hamiltonian. In my opinion, this is not descriptive, and serves little other than confusing the reader given the existing common usage of the term "S-matrix" elsewhere. A more boring but descriptive name, like "generalized bipartite theory", will likely be a better choice. Of course, in any case the authors have the jurisdiction on how they would want to call their theory.

Author Rebuttals to First Revision:

Reply to the Referee #2:

We are grateful to the referee for her/his appreciation of our updated version. Beyond her/his compliment, s/he also proposes to emphasize the relevance of the "S-matrix" in the material search by a one number indicator.

As the referee clearly knows, kagome, pyrochlore or Lieb lattices are *all* included in the S-matrix formalism, showing that it covers more than bipartite lattices (or slightly variant of them). Without diving into the accompanying theoretical paper (that will appear in Nature Physics), we provided in Appendix E1 how the kagome can be casted into the S-matrix formalism. Still, we understand the fair question of the referee: ***does this formalism bring anything beyond these three types mentioned above?***

The referee actually provides a very simple quantitative test: ***among the "best candidates" list, how many compounds host solely a bipartite lattice?*** As she/he pointed out, the flat band near a Fermi energy of a given compound hosting both a bipartite and another remarkable lattice (a Lieb, Kagome or pyrochlore) could arise from the bipartite sublattice and thus only captured by the S-matrix formalism. But such a systematic analysis is out of reach (as acknowledged by the referee). The number of compounds hosting only a bipartite lattice thus provides a ***lower bound*** for the number of compounds explained by the S-matrix formalism. Among the ***701*** ICSD entries in the "best candidates" list where we have detected at least one sublattice, ***155*** entries only host a bipartite lattice and thus can have their flat bands explained by the S-matrix formalism. So the yield (155/701 = 22%) is even better than the one quoted as "decent" (13%) proposed by the referee. We have added a few sentences in Appendix D3 to highlight this result.

In addition to this direct and concise answer to her/his suggested test, we would like to bring to the referee's attention two other quantitative pieces of evidence that the S-matrix formalism does provide an invaluable tool to probe flat bands, beyond the remarkable lattices.

1) In Appendix F and in the supplementary material of the accompanying theoretical article (in the version that will be published in Nature Physics), we have presented four prototypical examples of materials ($Pb_2Sb_2O_7$, WO_3, $CaNi_5$ and $Ca_2Ta_2O_7$) where the flat bands are captured by the S-matrix formalism and not by any other remarkable lattices. Overall, if we include materials with a similar structure, we cover 269 ICSD entries including 129 belonging to the "best candidates" list.

2) We would like to point out a recent preprint by C. Chui et al. arXiv:2109.13940 about line graph lattice crystal structures. They performed a high-throughput search of this specific type of lattice on our data set. Their results have been added to our database so we can test how their results overlap with our findings. The line graph lattices are also captured by the S-matrix formalism (note that the number 155 quoted above explicitly

excluded compounds with a line graph sublattice, including these latest would add 14 entries). Thus these results provide a concrete test to show that more flat band candidates are described by our approach than we actually promote in our own article.
Overall, this line graph high throughput search has found 1633 ICSD entries (related to 1044 unique materials) that are distinct from our own survey of bipartite, Lieb, split, kagome or pyrochlore lattices (i.e., the “None” category). Among them, 138 (15) ICSD entries belong to our list of curated (best) flat band candidates. Among them we have for example

- MgO (SG 223) https://www.topologicalquantumchemistry.fr/flatbands/index.html?ICSD=181465
- Al8Fe4Hf (SG 139) https://www.topologicalquantumchemistry.fr/flatbands/index.html?ICSD=607535
- Ge O3 Sr (SG 189) https://www.topologicalquantumchemistry.fr/flatbands/index.html?ICSD=28603

Even though this does not cover all the materials part of the “None” category, it clearly shows that another subclass of lattices encompassed by our S-matrix approach detects and captures potential candidate flat band materials not covered by the previous categories.
At the moment, there are no systematic approaches to find the S-matrix description of all flat band candidates. But we are confident that akin to the line graph example, specific subclasses of S-matrix models can be designed to largely complete this task, an information that would be then added to our database.

Finally, we understand the referee’s concern about the potential confusion when using the expression “S-matrix” in Physics where a similar term could mean the scattering matrix. Now other expressions such as “generalized bipartite theory” proposed by the referee also have their pitfalls. In that precise example, this would wrongly suggest that, e.g. line graphs or kagome lattices are excluded from this model which is not correct. Moreover, the term has already been coined and used in other publications (such as arXiv:2109.13940). For these reasons, we prefer to stick to the current denomination.